PROJECT DESIGN DOCUMENT

A Deep Stacked Ensemble Framework for Brain Abnormality Prediction Using Radiomics

1. Project Overview

The project is an MRI-based brain abnormality analysis framework designed around deep segmentation, ensemble mask generation, quantitative radiomics, neural feature learning, and visual explanation. The research notebook uses the BraTS 2020 training dataset, whose HDF5 slices contain four MRI channels and corresponding tumor masks. SegNet and a MONAI U-Net are trained to identify abnormal regions. Their prediction probabilities are averaged to obtain an ensemble mask. The ensemble mask is then used as the region of interest for PyRadiomics feature extraction. A compact fully connected neural network is trained over the extracted feature vectors, while Grad-CAM is used to highlight image regions that influence the segmentation network.

The packaged application is implemented with Streamlit. It accepts PNG, JPG, or JPEG MRI images, converts them to grayscale, resizes them to 512 × 512 pixels, runs a saved Dynamic U-Net model, overlays the predicted mask in red, and generates a Grad-CAM heatmap from the final convolutional layer. The source also loads a SegNet model, but the current interface does not call it during inference. Likewise, the saved radiomics neural model and feature CSV are not yet connected to the web application output.

2. Analyzed Project Components

Component	Role in the Project
final.py	Streamlit application, model loading, preprocessing, U-Net segmentation overlay, and Grad-CAM visualization.
model.py	A compact SegNet-style encoder-decoder accepting four input channels and producing a one-channel mask.
bts/model.py	Dynamic U-Net implementation with configurable filters and skip-connected encoder-decoder blocks.
bts/classifier.py	Training, testing, prediction, checkpointing, TensorBoard logging, Adam optimization, and Dice evaluation utilities.
bts/loss.py	Binary cross-entropy and Dice-based composite segmentation loss.
Notebook	Dataset download, metadata mapping, tumor-slice filtering, model training, ensemble mask generation, radiomics extraction, neural feature training, and Grad-CAM experiments.
features_df.csv	Stored radiomics/deep feature matrix produced from segmented tumor regions.
Model checkpoints	Saved weights for Dynamic U-Net, SegNet, and the radiomics neural model.
install.bat / req.txt	Environment installation and Python dependency definitions.

Table 1. Functional mapping of the supplied project files

3. Proposed System Architecture

The architecture is organized as a layered medical-image analysis pipeline. The input layer receives either BraTS HDF5 slices during experimentation or an uploaded image during deployment. The preprocessing layer standardizes spatial size and intensity representation. The segmentation layer contains complementary encoder-decoder networks. In the notebook, SegNet and MONAI U-Net operate on four-channel 256 × 256 inputs; in the Streamlit application, Dynamic U-Net operates on a single grayscale 512 × 512 input. The ensemble layer averages model probabilities and converts them into a binary region-of-interest mask. The radiomics layer computes quantitative descriptors from the masked region. The explainability layer generates Grad-CAM evidence. Finally, the application layer presents the original MRI, segmentation overlay, and heatmap.

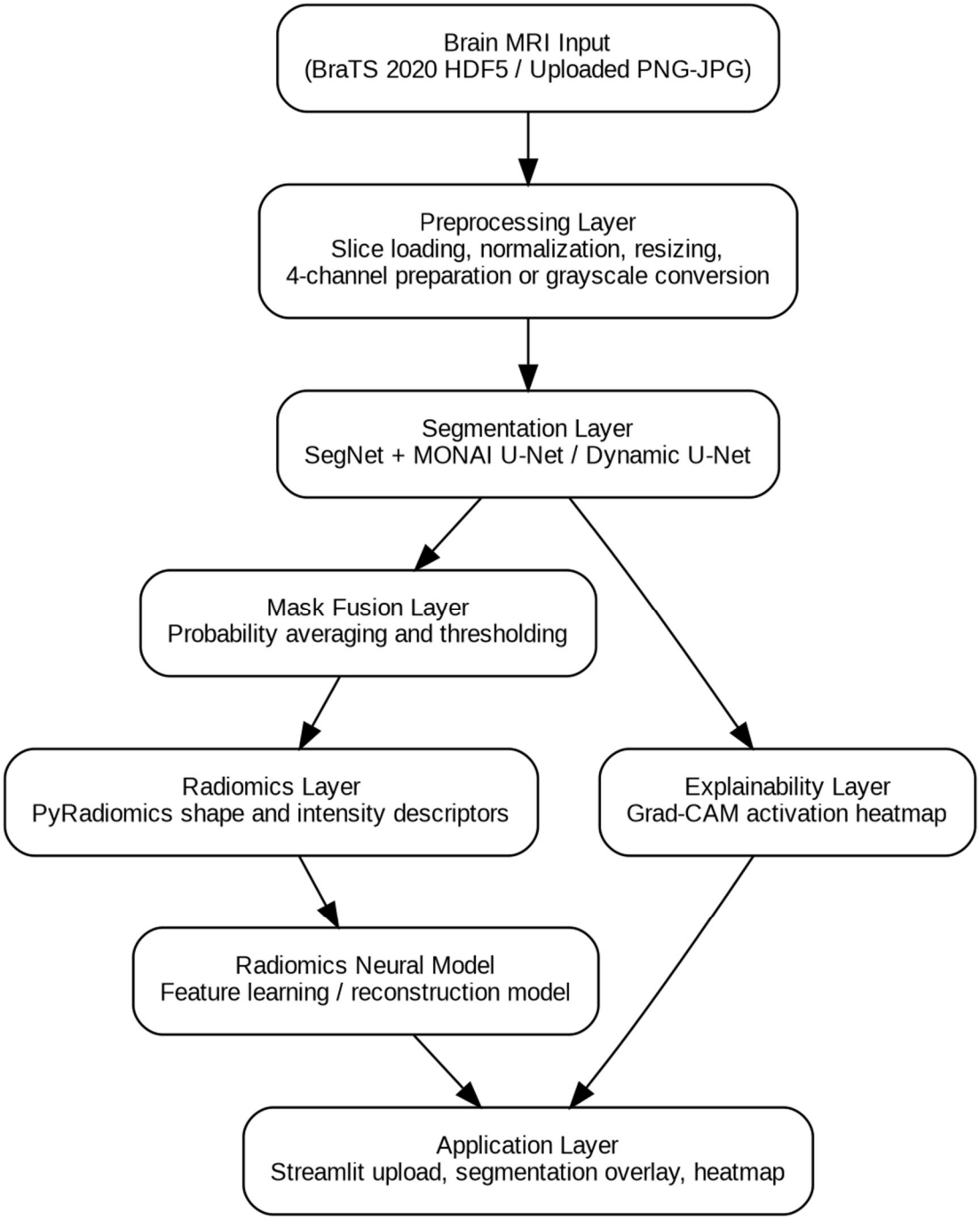


Figure 1. System architecture of the brain abnormality analysis framework

4. Project Workflow

The end-to-end workflow begins with data acquisition and finishes with interactive visualization. During model development, metadata is merged with subject-grade information, tumor-containing slices are selected, and the data are divided into training, validation, and test subsets. MRI channels are normalized and masks are binarized. SegNet and U-Net are optimized separately, after which their sigmoid probabilities are averaged. This ensemble mask defines the region from which radiomics descriptors are calculated. The resulting feature vectors are used to train a compact neural feature model. For interpretation, Grad-CAM backpropagates the summed segmentation response to a selected

convolutional layer and converts the weighted activations into a spatial heatmap. During deployment, the user uploads a scan and receives a red segmentation overlay and Grad-CAM visualization.

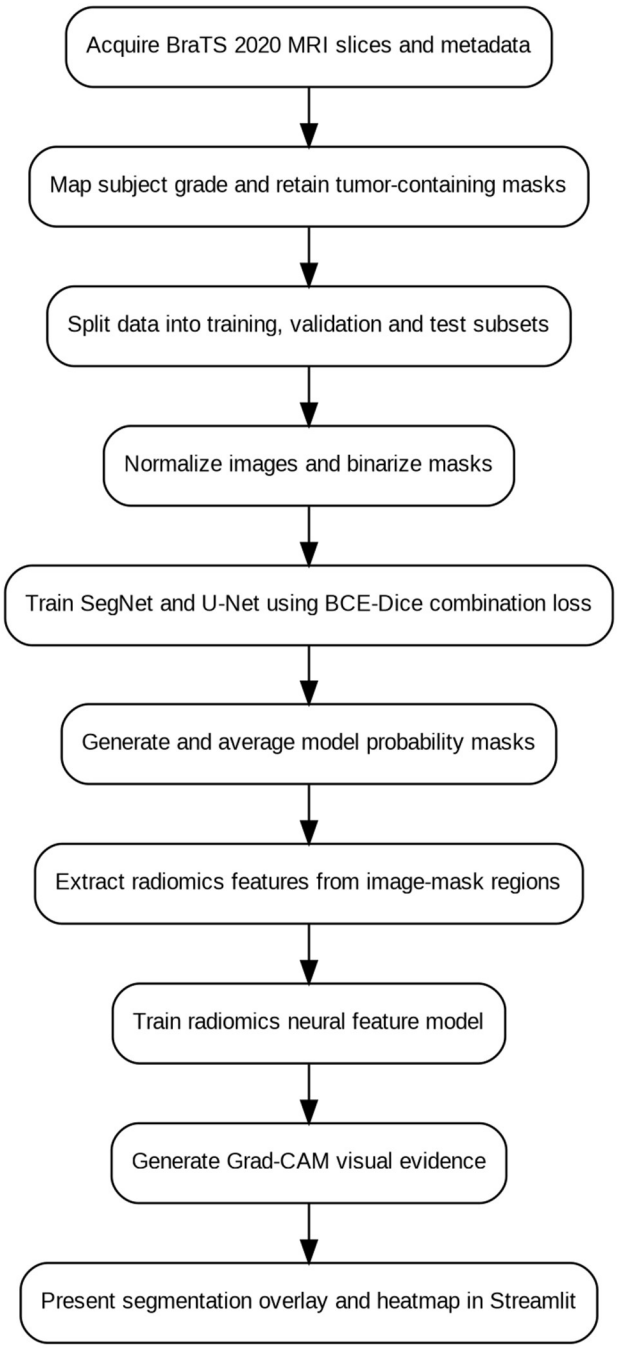


Figure 2. End-to-end workflow from MRI acquisition to visualization

5. Methodology

5.1 Dataset Preparation and Label Mapping

The notebook downloads the BraTS 2020 training dataset and reads MRI slices stored in HDF5 files. Each slice contains an image tensor and a segmentation mask. Metadata are merged with the name-mapping table to attach the tumor grade, where high-grade glioma and low-grade glioma are mapped to numeric classes. The implemented segmentation experiment limits loading to 2,000 slices and performs a 70% training, 15% validation, and 15% testing

division using a fixed random seed of 42. Additional filtering is used to avoid training on selections dominated by empty tumor masks.

5.2 Image and Mask Preprocessing

For notebook training, each MRI slice is prepared as a four-channel tensor and resized to 256 × 256 pixels. Intensity normalization uses the slice mean and standard deviation, with protection against near-zero variance. Ground-truth masks are converted to binary values so that all non-zero tumor labels are treated as abnormal tissue. The Streamlit inference path uses a different preprocessing configuration: the uploaded RGB image is converted to grayscale, resized to 512 × 512 pixels, and converted to a PyTorch tensor. This distinction must be preserved because the research and deployment paths currently use different channel and spatial configurations.

5.3 SegNet-Based Segmentation

The SegNet implementation contains two encoder stages and two decoder stages. The encoder uses 3 × 3 convolution, batch normalization, ReLU activation, and max pooling. Feature depth increases from 4 input channels to 64 and then 128 channels. The decoder performs bilinear upsampling, convolution, batch normalization, and ReLU before producing a one-channel segmentation logit map. A sigmoid operation is applied outside the model when probabilities are required.

5.4 U-Net-Based Segmentation

Two U-Net variants appear in the project. The notebook uses MONAI U-Net with four input channels, one output channel, channel widths of 16, 32, 64, and 128, residual units, and three downsampling strides. The application uses a saved Dynamic U-Net configured with filter widths [16, 32, 64, 128, 256]. The Dynamic U-Net uses encoder-decoder skip connections so that fine spatial information from early layers is combined with deeper semantic features during reconstruction of the tumor mask.

5.5 Composite Segmentation Objective

Training uses a combined Dice and binary cross-entropy objective. Binary cross-entropy provides pixel-wise supervision, while Dice loss directly promotes overlap between predicted and reference tumor regions. In the notebook, the combination is weighted toward Dice loss, which is appropriate for tumor segmentation because abnormal pixels occupy a much smaller area than background pixels.

$$L = \alpha\, L_Dice + (1 - \alpha)\, L_BCE, \quad \alpha = 0.6$$

5.6 Deep Ensemble Mask Generation

After separate inference by SegNet and U-Net, sigmoid converts their logits into pixel probabilities. The framework computes the arithmetic mean of the two probability maps. A threshold can then convert the fused probability map into a final binary mask. This ensemble reduces dependence on a single architecture and allows the complementary boundary behavior of the two encoder-decoder networks to contribute to the final region of interest.

$$P_ensemble(x) = [P_SegNet(x) + P_UNet(x)] / 2$$

5.7 Radiomics Feature Extraction

The ensemble mask is passed with the corresponding MRI image to PyRadiomics through SimpleITK. The extractor enables all available feature families and calculates image diagnostics, shape descriptors, and intensity/texture-related measures for the segmented region. The stored feature table also contains learned feature columns. Missing feature values are replaced with zero before conversion into PyTorch tensors. This stage transforms the segmented abnormality into a structured quantitative representation that can support prediction or representation learning.

5.8 Radiomics Neural Feature Model

A compact fully connected neural network is trained on the radiomics feature vectors. The notebook optimizes the network for 20 epochs using mini-batch learning. The saved radiomics_cnn_model.pth checkpoint represents this

feature-level model. Based on the supplied implementation, this component behaves as a neural feature-learning or reconstruction stage rather than a fully validated clinical class predictor, because the notebook does not provide a separate held-out classification report for brain abnormality classes.

5.9 Grad-CAM Explainability

Grad-CAM is attached to the final convolutional layer of the Dynamic U-Net. Forward and backward hooks capture activations and gradients. The total segmentation response is backpropagated, channel importance weights are computed by global averaging of gradients, and a rectified weighted combination of feature maps is resized and normalized. The resulting heatmap is color-mapped and blended with the original MRI to show where the network concentrated its response.

5.10 Streamlit Deployment

The Streamlit interface detects CPU or GPU availability, loads saved model weights, accepts a user image, and presents two analysis tabs. The segmentation tab runs the Dynamic U-Net, thresholds the output at 0.5, colors the mask red, and blends it with the scan. The Grad-CAM tab creates and overlays a heatmap. The interface is therefore suitable for demonstrating segmentation and explanation, but it does not currently display the SegNet-U-Net ensemble result, extracted radiomics values, tumor-grade output, or radiomics-model prediction.

6. Implemented Research Pipeline vs. Current Application

Function	Research Notebook	Streamlit Application
BraTS HDF5 loading and metadata mapping	Implemented in notebook	Not exposed
Four-channel MRI segmentation	Implemented in notebook	Uploaded images converted to one-channel grayscale
SegNet training and inference	Implemented and checkpoint stored	Model loaded but not called
U-Net segmentation	Implemented	Implemented with Dynamic U-Net
SegNet + U-Net ensemble	Implemented in notebook	Not exposed
PyRadiomics extraction	Implemented in notebook	Not exposed
Radiomics neural model	Trained and saved	Not loaded in final.py
Grad-CAM	Implemented	Exposed
Interactive visualization	Experimental plots	Exposed through Streamlit

Table 2. Alignment between the experimental and deployment implementations

7. Recommended Final Integrated Operation

For a complete system consistent with the project title, the application should use the same four-channel MRI representation as the trained ensemble, run both SegNet and U-Net, produce a fused mask, extract radiomics from that mask, and pass the resulting feature vector to a clearly defined supervised classifier. The user interface should then display the abnormality decision or tumor grade, confidence, fused segmentation, key radiomics descriptors, and Grad-CAM explanation. Model input dimensions, preprocessing, thresholding, and checkpoint architecture must be synchronized across training and deployment to prevent incompatible or misleading inference.

8. Conclusion

The supplied project demonstrates a substantial multi-stage brain MRI analysis workflow. Its strongest implemented elements are complementary deep segmentation models, probability-level mask fusion, radiomics extraction from the predicted region, neural feature learning, and visual explanation. The Streamlit application successfully demonstrates uploaded-image segmentation and Grad-CAM, but it currently represents only part of the notebook pipeline. The architecture and methodology documented here therefore describe both the completed experimental components and the exact integration work needed to turn them into one coherent brain abnormality prediction system.